

# Marvell Alaska 88E1310/88E1310S/88E1318/88E1318S

Integrated 10/100/1000 Mbps Energy Efficient Ethernet Transceivers

## PRODUCT OVERVIEW

Marvell® Alaska® 88E1310, 88E1310S, 88E1318 and 88E1318S Gigabit Ethernet (GbE) transceiver layer devices each containing a single Gigabit Ethernet transceiver. The transceiver implements the layer portion of the 1000BASE-T, 100BASE-TX, and 10BASE-T standards. The 88E1310S and 88E1318S support Synchronous Ethernet (SyncE) and Precision Time Protocol (PTP), which is based on IEEE 1588. The 88E1310 and 88E1318 support IEEE802.1AS.

The devices support RGMII (Reduced pin count GMII for direct connection) to Copper/Fiber/SGMII. The devices also integrate MDI interface termination resistors into the PHYs. This resistor integration simplifies board layout and reduces board cost by reducing the number of external components. The new MDI termination resistor scheme will achieve and exceed the accuracy requirements of the IEEE 802.3 return loss requirements.

The devices have an integrated switching voltage regulator to generate all required voltages and currents. The 88E1310/88E1310S support 2.5V/3.3V and the 88E1318/88E1318S support 1.8V/3.3V. The devices use advanced mixed-signal processing to perform equalization, channel cancellation, data recovery, and error correction at a gigabit per second data rate. They achieve high performance in noisy environments with very low power dissipation.

The Alaska family of transceiver products provides the ideal solution for rapid development and deployment of standalone and switching systems for the Enterprise, embedded, consumer, and Metro/service provider segments.

## APPLICATION BLOCK DIAGRAM

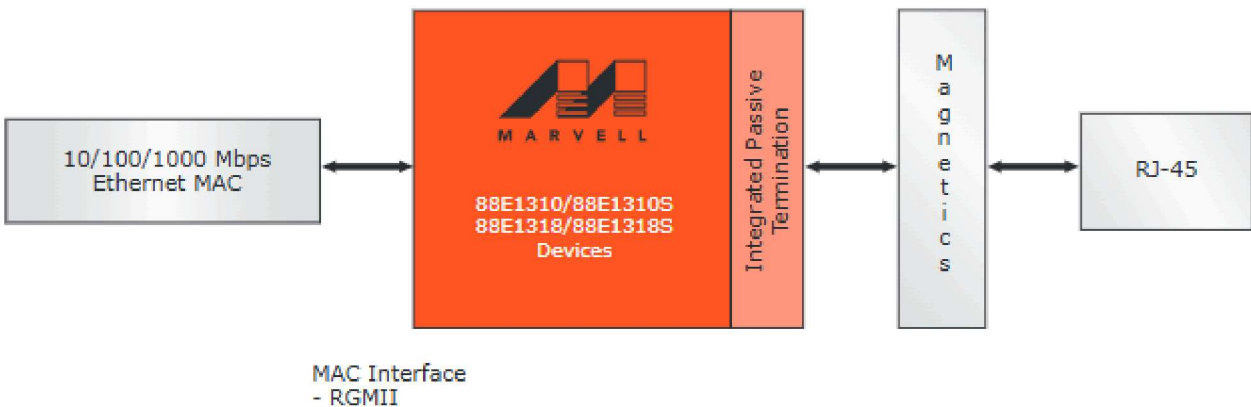


Fig 1. Alaska 88E1310/88E1310S/88E1318/88E1318S Application

## KEY FEATURES AND BENEFITS

| FEATURES                                                                                                                                                              | BENEFITS                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Four RGMII timing modes including integrated delays</li><li>IEEE1588 version 2 Time Stamping and Synchronous Ethernet</li></ul> | <ul style="list-style-type: none"><li>This eliminates the need for adding additional components</li><li>Enabling frequency and/or clock synchronization</li></ul> |